Probability
Lecture 8: Law of Large Numbers

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Content

1. Moment-Generating and Characteristic Functions
2. Law of Large Numbers
3. Practical applications

Moment-Generating and Characteristic Functions

- Recall that: for two independent random variables X and Y, let $Z = X + Y$

$$f_Z(z) = f_X * f_Y(z) = \int_{\mathbb{R}} f_X(x) f_Y(z-x) dx.$$

$\rightarrow$ tedious, hard to compute.

Definition

Given a random variable X, the **moment-generating function** (MGF) $M_X : \mathbb{R} \rightarrow \mathbb{R}$ is

$$M_X(s) = \mathbb{E}[e^{sX}].$$

- For discrete random variable,

$$M_X(s) = \sum_x e^{sx} p_X(x).$$

- For continuous random variable, (Laplace transformation)

$$M_X(s) = \int_{\mathbb{R}} e^{sx} f_X(x) dx$$

Moment-Generating and Characteristic Functions

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Example

Find the moment-generating function of the following random variables:

- X with three states $0, 1, 2$ with probability masses $2/6; 3/6; 1/6$.
- $Poi(\lambda)$: $\mathbb{P}(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}$.
- $Exp(\lambda)$: $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$.

Example

- X with three states $0, 1, 2$ with probability masses $2/6; 3/6; 1/6$.

$$M_X(s) = \mathbb{E}[e^{sX}] = e^{s.0}\frac{2}{6} + e^{s.1}\frac{3}{6} + e^{s.2}\frac{1}{6} = \frac{1}{3} + \frac{e^s}{2} + \frac{e^{2s}}{6}.$$

- $Poi(\lambda)$: $\mathbb{P}(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}$.

$$M_X(s) = \mathbb{E}[e^{sX}] = \sum_{k=0}^{\infty} e^{sx} e^{-\lambda}\frac{\lambda^k}{k!} = e^{-\lambda}\sum_{k=0}^{\infty}\frac{(\lambda e^s)^k}{k!} = e^{-\lambda}e^{\lambda e^s}.$$

- $Exp(\lambda)$: $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. For $\lambda > s$

$$M_X(s) = \mathbb{E}[e^{sX}] = \int_0^{\infty} e^{sx}\lambda e^{-\lambda x} dx = \frac{\lambda}{\lambda - s}$$

Properties

Theorem

The moment-generating function has the properties

- $M_X(0) = 1$.
- $\left.\frac{d}{ds} M_X(s)\right|_{s=0} = \mathbb{E}[X], \quad \left.\frac{d^2}{ds^2} M_X(s)\right|_{s=0} = \mathbb{E}[X^2]$.
- $\left.\frac{d^k}{ds^k} M_X(s)\right|_{s=0} = \mathbb{E}[X^k]$ *for any* $k \in \mathbb{N}$.

Distribution table

Distribution	PMF / PDF	$\mathbb{E}[X]$	$\mathrm{Var}[X]$	$M_X(s)$
Bernoulli	$p_X(1) = p$ and $p_X(0) = 1-p$	p	$p(1-p)$	$1-p+pe^s$
Binomial	$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$	np	$np(1-p)$	$(1-p+pe^s)^n$
Geometric	$p_X(k) = p(1-p)^{k-1}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$\frac{pe^s}{1-(1-p)e^s}$
Poisson	$p_X(k) = \frac{\lambda^k e^{-\lambda}}{k!}$	λ	λ	$e^{\lambda(e^s-1)}$
Gaussian	$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$	μ	σ^2	$\exp\left\{\mu s + \frac{\sigma^2 s^2}{2}\right\}$
Exponential	$f_X(x) = \lambda \exp\{-\lambda x\}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\frac{\lambda}{\lambda - s}$
Uniform	$f_X(x) = \frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{sb} - e^{sa}}{s(b-a)}$

Figure: Moment-generating functions of common random variables

Sum of independent variables via MGF

Theorem

- Let X and Y be independent random variables. Let $Z = X + Y$. Then

$$M_Z(s) = \mathbb{E}[e^{sZ}] = \mathbb{E}[e^{s(X+Y)}] = \mathbb{E}[e^{sX}e^{sY}] = M_X(s)M_Y(s).$$

- Consider independent random variables $X_1, X_2, \ldots, X_n$. Let $Z = X_1 + \ldots + X_n$. Then

$$M_Z(s) = \prod_{i=1}^{n} M_{X_i}(s).$$

Theorem

- Sum of indep. Bernoulli = Binomial. (same p)
- Sum of indep. binomial = binomial. (same p)
- Sum of indep. Poisson = Poisson.
- Sum of indep. Gaussian = Gaussian

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Characteristic functions

Definition

The **characteristic function** of a random variable X is

$$\varphi_X(t) = \Phi_X(it) = \mathbb{E}[e^{-itX}].$$

$\rightarrow$ Fourier transformation.

Fourier Transforms $f(t) \longleftrightarrow F(\omega)$

1.	$e^{-at}u(t) \longleftrightarrow \frac{1}{a+j\omega}$, $a>0$	10.	$\text{sinc}^2(\frac{Wt}{2}) \longleftrightarrow \frac{2\pi}{W}\Delta(\frac{\omega}{2W})$
2.	$e^{at}u(-t) \longleftrightarrow \frac{1}{a-j\omega}$, $a>0$	11.	$e^{-at}\sin(\omega_0 t)u(t) \longleftrightarrow \frac{\omega_0}{(a+j\omega)^2+\omega_0^2}$
3.	$e^{-a\lvert t\rvert} \longleftrightarrow \frac{2a}{a^2+\omega^2}$, $a>0$	12.	$e^{-at}\cos(\omega_0 t)u(t) \longleftrightarrow \frac{a+j\omega}{(a+j\omega)^2+\omega_0^2}$
4.	$\frac{a^2}{a^2+t^2} \longleftrightarrow \pi a e^{-a\lvert\omega\rvert}$, $a>0$	13.	$e^{-\frac{t^2}{2\sigma^2}} \longleftrightarrow \sqrt{2\pi}\sigma e^{-\frac{\sigma^2\omega^2}{2}}$
5.	$te^{-at}u(t) \longleftrightarrow \frac{1}{(a+j\omega)^2}$, $a>0$	14.	$\delta(t) \longleftrightarrow 1$
6.	$t^n e^{-at}u(t) \longleftrightarrow \frac{n!}{(a+j\omega)^{n+1}}$, $a>0$	15.	$1 \longleftrightarrow 2\pi\delta(\omega)$
7.	$\text{rect}(\frac{t}{\tau}) \longleftrightarrow \tau\text{sinc}(\frac{\omega\tau}{2})$	16.	$\delta(t-t_0) \longleftrightarrow e^{-j\omega t_0}$
8.	$\text{sinc}(Wt) \longleftrightarrow \frac{\pi}{W}\text{rect}(\frac{w}{2W})$	17.	$e^{j\omega_0 t} \longleftrightarrow 2\pi\delta(\omega-\omega_0)$
9.	$\Delta(\frac{t}{\tau}) \longleftrightarrow \frac{\tau}{2}\text{sinc}^2(\frac{\omega\tau}{4})$	18.	$f(t)e^{j\omega_0 t} \longleftrightarrow F(\omega-\omega_0)$

Example

- Let X and Y be independent, and let

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases} \quad \text{and } f_Y(y) = \begin{cases} \lambda e^{-\lambda y}, & y \geq 0 \\ 0, & y < 0 \end{cases}$$

Find the PDF of $Z = X + Y$.

$$\varphi_X(t) = \varphi_Y(t) = \frac{\lambda}{\lambda + jt},$$

$$\varphi_Z(t) = \varphi_X(t).\varphi_Y(t) = \frac{\lambda^2}{(\lambda + jt)^2}.$$

Then, $f_Z(z) = \lambda^2 z e^{-\lambda z}$ for $z \geq 0$.

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1 Moment-Generating and Characteristic Functions

2 Law of Large Numbers

3 Practical applications

Motivation

- Sample data $\{X_1, \ldots, X_N\}$ **i.i.d** (independent identically distribution)
- Sample average

$$\overline{X} = \frac{X_1 + X_2 + \ldots + X_N}{N}.$$

- Expectation

$$\mathbb{E}[\overline{X}] = \frac{\mathbb{E}[X_1] + \mathbb{E}[X_2] + \ldots + \mathbb{E}[X_N]}{N} = \mu.$$

- Variance

$$\begin{aligned}\mathrm{Var}(\overline{X}) =& \frac{\mathrm{Var}(X_1 + X_2 + \ldots + X_N)}{N^2} \\ =& \frac{\mathrm{Var}(X_1) + \ldots + \mathrm{Var}(X_N)}{N^2} = \frac{\sigma^2}{N} \to 0\end{aligned}$$

Motivation

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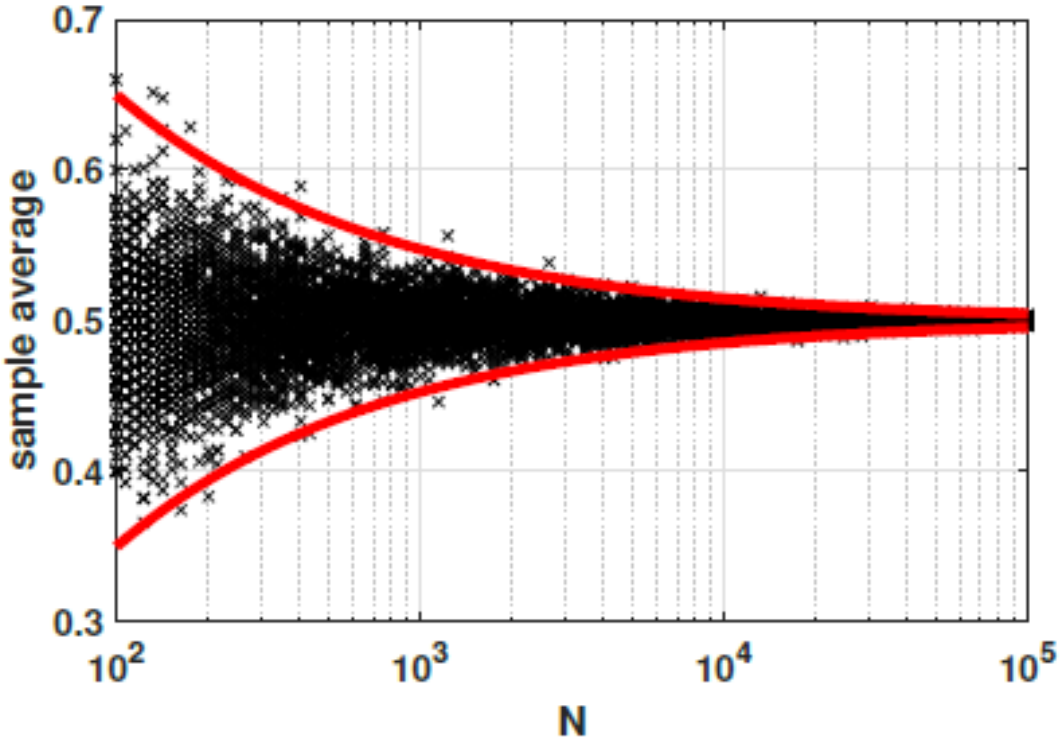
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0.7
0.6
0.5
0.4
0.3
sample average
10^2
10^3
10^4
10^5
N

Weak law of large numbers (WLLN)

Theorem

Let $X_1, \ldots, X_N$ be a set of i.i.d random variables with mean μ and variance σ^2. Then for every $\epsilon > 0$,

$$\lim_{N \to \infty} \mathbb{P}[|\overline{X} - \mu| > \epsilon] = 0.$$

How to prove?

- Markov inequality: Let $X \geq 0$

$$\mathbb{P}[X \geq \epsilon] \leq \frac{\mathbb{E}[X]}{\epsilon}.$$

-Chebyshev inequality:

$$\mathbb{P}[|X - \mathbb{E}X| \geq \epsilon] = \mathbb{P}[|X - \mathbb{E}X|^2 \geq \epsilon^2] \leq \frac{\mathbb{E}[(X - \mathbb{E}X)^2]}{\epsilon^2} = \frac{\mathrm{Var}(X)}{\epsilon^2}.$$

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Convergence in Probability

Definition

A sequence of random variables $X_1, \ldots$ **converges in probability** to a deterministic number α if for every $\epsilon > 0$,

$$\lim_{N \to \infty} \mathbb{P}[|X_N - \alpha| > \epsilon] = 0.$$

Denote $X_N \xrightarrow{\mathbb{P}} \alpha$.

Example

- Let $X_1, \ldots, X_N$ be i.i.d. random variables with $Uniform(0,1)$. Define $A_N = \min(X_1, \ldots, X_N)$. Show that A_N converges in probability to zero. + For every $\epsilon > 0$,

$$\mathbb{P}(A_N > \epsilon) = \mathbb{P}(\min(X_1, \ldots, X_N) > \epsilon) = (1-\epsilon)^N \to 0.$$

- Let $X \sim Exp(1)$, i.e $\mathbb{P}[X \geq x] = e^{-x}x$. Let $A_N = X/N$. Prove that A_N converges to zero in probability. + For every $\epsilon > 0$,

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Strong law of large numbers (STLN)

Definition

A sequence of random variables $X_1, \ldots$ **converges almost surely** to a deterministic number α if

$$\mathbb{P}[\omega \in \Omega : \lim_{N \to \infty} X_N(\omega) = \alpha] = 1.$$

Denote $X_N \overset{a.s}{\to} \alpha$.

Theorem

Let $X_1, \ldots, X_N$ be a set of i.i.d random variables with mean μ. Then

$$\overline{X} \overset{a.s}{\to} \mu.$$

1 Moment-Generating and Characteristic Functions

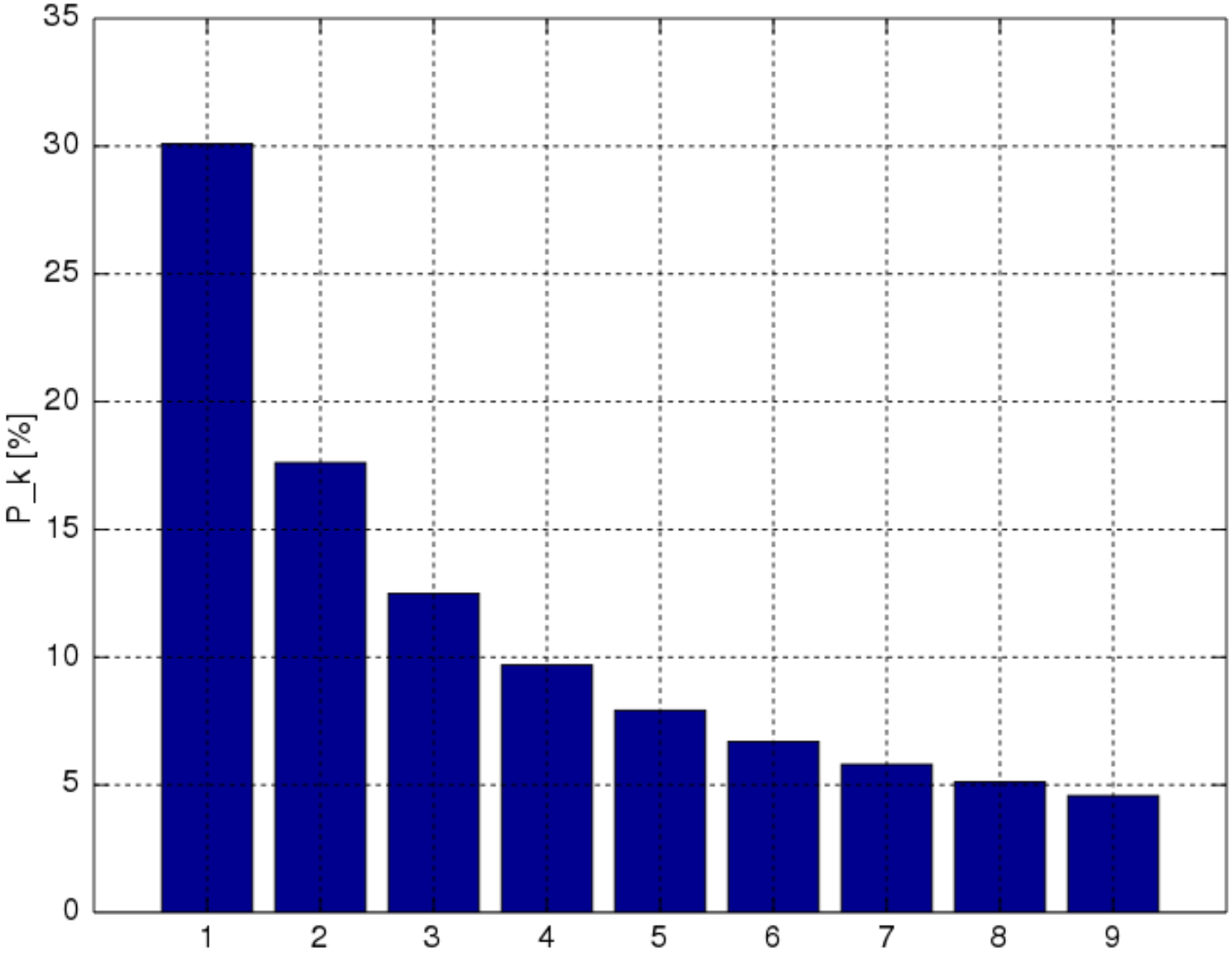
2 Law of Large Numbers

3 Practical applications

Point estimation

- Aim: estimate a parameter through expectation (or probability).
- Probability:
  - Poll election, Birth rate,...
  - Check fair coin
  - Buffon needlet problem: estimate the number π
- Expectation: average height, weight, income, lottery...

Benford law



Numerical calculation an integral

- Consider an integral, for example

$$\int_0^1 f(x)dx.$$

- Hard to compute exactly.
- Idea: let $U \sim Unif[0,1]$

$$\mathbb{E}[f(U)] = \int_0^1 f(x)dx$$

- Use LLN to estimate $\mathbb{E}[f(U)]$: generate randomly $U_1, \ldots, U_n$

$$\frac{f(U_1) + f(U_2) + \ldots + f(U_n)}{n} \simeq \mathbb{E}[f(U)] = \int_0^1 f(x)dx.$$

$\rightarrow$ MONTE-CARLO method!!!

Weierstrass approximation theorem

- Theorem: any continuous function on $[0,1]$ can be approximated by a polynomial.
- Idea:
  - Consider the binomial distribution $X \sim B(n,x)$ for $x \in [0,1]$
  - By LLL
$$\frac{X}{n} \to x \quad (\mathbb{P},\ a.s).$$
  - Hence $f(X/n) \to f(x)$ and $\mathbb{E}[f(X/n)] \to f(x)$.
  - Note that (Bernstein polynomial)
$$\mathbb{E}[f(X/n)] = \sum_{k=0}^{n} f(k/n) C_n^k x^k (1-x)^{n-k}.$$

Homework

Chan: Chapter6. 1-7, 16-18.

Thank you!